

Feeling Data through Movement: Designing Somatic Data Experiences with Dancers

Laura J. Perovich
l.perovich@northeastern.edu
Northeastern University
Boston, MA, USA

Nicole Zizzi
n.zizzi@northeastern.edu
Northeastern University
Boston, MA, USA

ABSTRACT

The proliferation of data throughout society has produced an increased interest in novel ways to represent information to provide new insights, re-humanize data, or allow broader participation. Many novel data representations allow people to physically interact with information by picking up, touching, and manipulating artefacts that encode data through their material properties. In this paper, we draw on the arts to explore how we might design for deeper physical engagement with data. How can we feel data through movements, instead of through objects? We examine nine instructional movement sequences created by five dancers as part of a larger workshop on data and dance. Our findings shed light on promising design strategies for somatically-focused data representations including observations on the relational, layered, and emotionally engaged approaches taken by dancers. We also reflect on these approaches in relation to soma design and identify areas for future work in "data somatics."

CCS CONCEPTS

• **Human-centered computing** → **Interaction techniques**; **Visualization**.

KEYWORDS

novel data representations, somatics, embodiment, dance

ACM Reference Format:

Laura J. Perovich and Nicole Zizzi. 2024. Feeling Data through Movement: Designing Somatic Data Experiences with Dancers. In *Eighteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '24)*, February 11–14, 2024, Cork, Ireland. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3623509.3633371>

1 INTRODUCTION

The importance of data in society continues to increase as it becomes more abundant and is used in decision-making processes that impact our everyday lives. Because of the broadening audience for data—and its connections to power and policy—researchers, designers, and communities are exploring novel approaches to interacting with data that serve diverse audiences, needs, and aims. Much of this research goes beyond screen-based visualization to

include physical, multi-sensory, or multi-media information representations. Subfields like data physicalization [40], data sculpture [63], and data visceralization [15, 16] speculate about potential benefits of increased accessibility and engagement with the important social issues data attempt to quantify.

Researchers have speculated that some of the possible benefits of novel data representations are connected to the physical nature of these alternative representations and people's ability to use their bodies to interact with them in the physical world [40]. Some of these approaches build from tangible interfaces research [43, 46, 53] and embodied cognition which considers the body and objects around the body to be an extended space for cognition [54]. Recently, embodied approaches to data representation have drawn on theater and dance to perform data through bodies [7, 13]. These approaches take the important step of centering bodies in motion as a way to represent information. Yet many people's experience of the information is still visual—the works are designed to be performed for a viewing audience and a comparatively small number of people have the embodied experience of enacting the piece.

In this paper, we center the possibility of body-based interactions with data by removing the artefact from the experience and by moving, not watching; we design experiences of data that start and end with(in) the body. We explore the design space for somatic representations of data through activities with expert practitioners. Specifically, we analyze outcomes from three activities where dancers designed movement sequences to represent data as part of a larger workshop series on data and dance. Five dancers participated and designed nine movement sequences that they enacted and reflected on. Dancers were specifically instructed to create movement sequences that would allow people to "feel" the data in their body through doing—not performance-based works that would be observed by an audience. In that sense, our efforts dovetail with work in soma design that begins from—and seeks to create—a rich, felt experience [29, 60].

We analyze these nine movement sequences primarily through the lens of design, considering what strategies expert movers bring to bear on this task to shed light on the possibilities in this space. Our findings surface the relational, layered, and emotionally engaged approaches taken by dancers and also consider the dancers' overall experience with these somatic methods of data representation. In the discussion, we reflect on the ways that somatically-focused representations of data—or "data somatics"—might expand, align with, complement, or supplement existing data display practices. We end by suggesting future research directions based on our explorations.



This work is licensed under a Creative Commons Attribution-NonCommercial-ShareAlike International 4.0 License.

TEI '24, February 11–14, 2024, Cork, Ireland
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0402-4/24/02.
<https://doi.org/10.1145/3623509.3633371>

2 RELATED WORK

Our exploration of embodied approaches to creating a felt experience of data in the body relates to a number of ongoing research threads in TEI. We begin with a summary of novel approaches to data representation that focuses on work from the HCI and visualization communities and highlights body-based efforts in these areas. We then consider body-based approaches more broadly and describe the ways in which the "somatic turn" in HCI can inform our efforts to create movement and body-based data representations. Finally, we describe how our work builds from, complements, and extends existing research in these fields.

2.1 Novel and body-based data representations

Interest in novel data representations has increased as data has become more important and prevalent in our daily lives. Communities and artists have long worked in these areas and have created many physical or embodied ways of interacting with information from pebble-based voting systems [38] to string-based diaries of life events [2] to performative knitting of sleep data [3]. In 2010, artist Kelly Dobson proposed the area of data visceralization where data is physically experienced instead of being extracted, separated from the body, and controlled through a digital database [15, 16]. This proposition creates space for a whole-body multi-sensory feeling of information in data representation and takes an integrated view of bodily experience, understanding, and emotion.

Data physicalization has been a center point of work on novel data representations within the HCI and data visualization communities. Researchers speculate that physical representations of data may afford a number of possible benefits in certain situations by, for example, increasing accessibility [8, 25, 36, 40], improving information recall [57], increasing efficiency in information retrieval tasks [39], and creating space for emotional engagement [48, 62] compared to traditional visualizations. Data physicalizations have been created based on a wide variety of data, media, and aims. Most data physicalizations are objects. The way these objects are interacted with or the embodied experiences they create for a person are typically not the primary concern and may not be directly considered in the design.

At the same time, some projects are more intentional about interactions with data representations and the feelings those interactions might facilitate. The H3 (Hydrogen cubed) and the Solar Radiation Dowsing Rod projects are examples of haptic approaches to data physicalization [26]. In both of these projects people interact with a handheld object to get information which is communicated through vibration; specifically, stronger vibrations indicate higher levels of hydrogen in that location (H3) or higher levels of solar radiation in the direction indicated by a rod (Solar Radiation Dowsing Rod). The vibration also creates a sound which reinforces the haptic experience. Karyda et al. provide examples of movement-centered experiences that can be facilitated by physical representations of data. Body-based interactions with data are brought to the forefront through musical instruments, table top games, and exercise equipment that represent personal data through the interaction itself [42]. Overall, these projects have limited visual modes of representation—the data is primarily understood through the physical (or sonic) experience imparted by the objects.

Wearable representations of data may also be used to bring information closer to the body. Some wearable data displays primarily use visual methods to convey information, such as knitted shrugs with heart rate data [41] and screen printed shirts with data on household pollutants used in environmental health report-back [47]. Researchers and study participants note that wearing information on the body can create space for reflection on personal data and surface opportunities for conversations with others—opportunities that may not emerge with traditional visualizations [41, 47]. Other wearables use dynamic bodily sensations to represent data. Embodisuit uses actuators to represent data through heat and vibrations [11], and WARMTH draws on the embodied experience of body heat from physical contact and uses temperature in a wearable to represent different distance between family members [6]. Overall, placing data in garments situates it on (or near) the body and creates space to begin incorporating felt sensations in the design.

Recently, researchers have proposed performative or movement oriented approaches to data representation that they call "human data embodiment" [37] or just "data embodiment" [13]. Early examples from this still emerging space draw on theater and dance to represent information through movement, action, and interaction to create a performance of data for an audience. In 100% City by the theater company Rimini Protokoll, people represent demographic data in a staged performance—that is, "living human bodies" are the "material for data physicalization" [37]. Actors answer questions and move around the stage to help the audience gain an understanding of a city which "give[s] statistics a face" [37]. In Data Sensorium, Derry takes inspiration from dance to represent environmental migration data through movement and graphics [13]. By using gestures to represent different disasters and movements through space to encode the number of people displaced, the piece creates a kinetic and somatic experience for the dancers—and a performance that can be viewed by an audience.

2.2 Embodied and somatic approaches within HCI

Our work in this paper reflects an ongoing interest in embodied and somatic approaches in HCI. Lian Loke's and Thecla Schiphorst's 2018 Interactions paper declared the dawn of the "somatic turn" in HCI and celebrated researchers' attention to the importance of considering rich embodied experience [45]. These somatic approaches build from Western movement practices such as Feldenkrais Method [20], Bartenieff Fundamentals [5, 24], and Viewpoints [10]—which themselves build on Eastern movement practices such as Yoga, Tai chi, and Qigong [19]—to center bodily sensation, and cultivate self-awareness. They also expand our understanding of what HCI includes and dovetail with a long term effort to identify "where the action is" for the technologies we design [17, 18].

During this time, soma design evolved into an important area of work within HCI. Soma design uses bodymind approaches to both iteratively design an artefact based on felt experiences and build one's own soma—that is, one's sensing abilities and embodied knowledge [56]. In this active practice, doing and moving are valued [60]—researchers are advised to "stop talking – start feeling" [29] because this "allows the felt experience to take the front seat" [29]. Soma design is an embodied individual and particular "first-person"

experience [14, 30, 56] and also part of a larger effort towards collective awareness, well-being, and ethics [28]. It has been applied to a number of topics, including mindfulness [55], body augmentation [56, 58], everyday household products [44, 60], sexuality [34], and health [12] and leveraged relative to technologies such as virtual reality [4, 59] and wearables or textiles [32, 34, 56].

Soma design methods have also been used around biodata and in biosensing applications. This includes work with data on breath [23, 61], heartbeat [4, 41], and emotion [33], among others. The artefacts resulting from these design processes take a number of forms, include micro drones controlled by breathing [23] and real-time sonic representations of contractions of a person's calf muscle as they walk along a beam in (and out of) virtual reality [59]. Overall, these projects aim to help people examine and expand their sense of embodiment and their relationships with their bodies through somatic experiences around biological data. These approaches stand in contrast to self-tracking methods that center health outcomes or seek to "optimize" people's bodies. Instead, they move towards a "somadata" approach that helps people connect biodata directly to their felt bodily experiences [1].

Experts in embodiment—or "somatic connoisseurs"—are a key part of soma design. Somatic practitioners, dancers, actors, or others with embodied training may be invited to participate as members of the design team or facilitators [31, 52, 60]. Somatic connoisseurs contribute an embodied expertise that is unfamiliar to most designers and can expand the design space and help the team become more aware of their bodies [60]. Other HCI researchers have also drawn inspiration from somatic practices, including work on physicalization and musification based in the Alexander Technique [49].

This turn towards embodiment has also gained some traction in data visualization, particularly through data feminism. A core principle of data feminism is to "elevate emotion and embodiment" [15]. This principle highlights the importance of knowledge that comes from our lived experience as sensing bodies in the world. Data feminism builds from data visceralization and encourages multisensory data experiences that use sound, touch, or taste in representations [15]. Manifestations of this principle include performances of the representation of female artists in the MoMA collection and edible pie-based pie charts about gender equality [15, 35, 50].

2.3 Summary

HCI and visualization research has an ongoing interest in novel ways of displaying information and an emerging focus on somatics for design. Data physicalizations largely focus on creating artefacts that encode data through their material properties. Arts-based or emerging approaches such as data embodiment and data visceralization consider what it means to represent data with and through the human body, often using performance or multisensory methods.

Our work in this paper draws from this history of novel data representation. It expands on this prior work by bringing a new focus to the somatic—the potential for a felt experience of data. How can we sense data through embodied experience, not objects? What is it to design embodied movement-based data representations that are only for feeling, not for viewing? Inspired by soma design, we engage somatic connoisseurs—dancers—to explore these questions and suggest how we might begin to design in this space.



Figure 1: Dancers participated in three instructional activities described in this paper as part of a workshop series about data and dance. In these activities, dancers acting as "designers" created instructional movement sequences to convey a felt experience of data through motion. Dancers acting as "movers" enacted the movement sequences by following the designers' verbal ("verbal", left) or physical ("flocking", right) instructions and reflected on their embodied experiences.

3 METHODS

3.1 Workshops and participants

The activities described in this paper were part of a series of ten 1.5 hour workshops conducted in the summer of 2022. Most workshops were held as double sessions to accommodate the dancers' schedules. The workshops were designed and led by the paper's authors who each have over ten years of experience in dance and somatic practices.

Five dancers participated in at least one of the instructional movement activities described in this paper; an additional dancer participated in many of the workshops, but was not present for these sessions. All five dancers are active members of a modern or contemporary dance group, and four of the dancers are current members of a company co-led by one of the authors. Two of the five dancers have experience choreographing and one dancer's primary career is in dance education. Four had been dancing since childhood and many had experience in other styles such as ballet. Dancers had a range of comfort with and understanding of data prior to the workshops. Three dancers reported working with data regularly in their profession or education, one dancer reported limited prior experience with data, and one dancer was between these two groups. All dancers identify as female or non-binary.

3.2 Movement activities

The movement sequences described in this paper (Table 1) were a result of activities in workshops 7, 9, and 10. In these activities, dancers were asked to design instructions that would guide people through a series of movements that would allow them to feel—not see—a particular set of data. That is, these pieces would be done—not viewed—by the "audience."

For each activity, dancers acted as "designers" and "movers" (Figure 1). In the designer role, dancers selected the data they wanted to work with from a predetermined set of simple visualizations (Figure 2), created a set of instructions, and led the group through the movement sequence three times. In the "movers" role, dancers followed the instructions given by the designers. Two activities used a verbal approach to conveying the instructions; that is, the designers told the movers what to do in real-time without physical demonstration

Table 1: Nine instructional movement sequences are included in the analysis. Movement parameters in italics were not explicitly described by the designers, but were observed by the movers or the researchers.

Activity	Movement sequence	Designers	Activity type	Data visualization	Movement parameters	Movement sequence type
Instructional activity #1 (workshop 7)	M01	P01	verbal	bar graph	effort, height, emotion	technical
	M02	P02	verbal	bar graph	weight, effort, height	imagery
	M03	P03	verbal	bar graph	effort, weight	imagery
	M04	P04	verbal	bar graph	effort, <i>emotion</i> , <i>space</i>	narrative
	M05	P05	verbal	bar graph	height, <i>effort</i> , <i>weight</i>	technical
Instructional activity #2 (workshop 7)	M06	P01, P04, P05	flocking	line graph	height, direction, <i>shape</i>	technical
	M07	P02, P03	flocking	line graph	height, shape, direction	technical
Instructional activity #3 (workshop 9-10)	M08	P05, P02	verbal	line graph	height, direction, <i>time</i> , <i>effort</i> , <i>weight</i>	technical
	M09	P01, P04	verbal	line graph	height, scale	technical

of the movement. One activity used a "flocking" approach to the instructions; that is, the designers led the group through the movements by doing—similar to flocking birds. Sometimes designers supplemented flocking with limited verbal instructions.

These activities were loosely inspired by instruction-based approaches of artists such as Sol Lewitt (wall drawings), Simone Forti (Dance Constructions), and John Cage (Happenings) that specify a framing while also creating space for the variability that emerges from an individual hand, a moment in time, and the divergence of embodied experience. The larger workshop series—which will be reported on in future papers—included a variety of activities that explored movement parameters¹ and different ways to express data through and with the body in motion.

3.3 Data and data analysis

All workshop activities were video and audio recorded. Dancers were asked to discuss their experience during each activity. These semi-structured discussions explored what the movers felt, what stood out to them about the movement, and what they thought the data was like based on their experience. Movement designers were asked about the intent of their design, what movement parameter(s) they used to represent the data, and what the original data visualization looked like. The audio from these semi-structured discussions were transcribed; some portions of the audio recordings were difficult to capture due to noise in an adjacent dance studio. Artefacts from the design process—such as notes and movement instructions—were photographed. At the end of each workshop, dancers provided written responses to six open-ended questions asking them to describe their experience at the workshop and their process, reflect on what went well and what was challenging, and ask any questions that they had. They also completed individual semi-structured interviews at the end of the workshop series to reflect on their overall experience. Workshop leaders (the paper

authors) debriefed together after each session to discuss what they had observed, describe salient moments, and reflect on possible research implications. Debriefing conversations and interviews were recorded and transcribed.

Data analysis began by both authors collecting and reviewing all the data related to the activities (described above). The authors then iteratively identified a few core questions to guide the analysis: (1) what fundamental movement parameters and structures do dancers use to design for felt experience? (2) what other strategies do the dancers use to design for felt experience? (3) what are the dancers' experiences of designing for felt experience?.

Each author led the analysis of about half of the movement sequences based on these guiding questions and using a template in a shared document. Researchers synthesized data across sources to create an annotated visualization of each movement sequence that unpacked its components, encodings, and representation strategies. The document also included comments from the dancers and observations from the researchers captured during and after the workshops. Researchers discussed preliminary findings with each other weekly throughout the analysis process and exchanged movement sequences to iteratively converge to a final set of findings.

Overall, this analysis aimed to identify design opportunities that engage the felt (not visual) experience when representing data. As this is one of the first fully somatic movement-based approaches to data representation in HCI, it intends to provide a starting point for future work without claiming to be an exhaustive account of what is possible.

4 FINDINGS

Our findings aim to begin to map out this space; we expect new findings to emerge as this research continues with different groups, data, and activities. We begin by identifying the basic representation strategies used in the movement sequences. We then describe multiple ways movements are layered in data representations, how emotion was incorporated, and relational approaches used and experienced by the dancers. Finally, we consider feedback from the dancers about these approaches to data representation.

¹We use the phrase "movement parameters" to describe the dimensions along which people can intentionally vary their movements. For example, a dancer might alter the height of their movement or the effort they exert while doing a movement. Dancers in the workshops were already familiar with frameworks for movement from their past training and we continued to discuss them throughout the workshops.

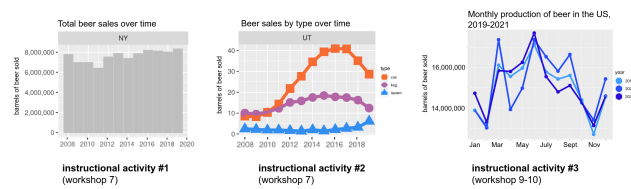


Figure 2: Sample data visualizations used by the dancers to design instructional movement sequences for each activity. Dancers selected one visualization to design from out of a number of similar options—for example, in instructional activity #1 they were offered visualizations from each of the 50 United States. Example movement sequences for the visualizations shown here are included in Figures 3 - 5

4.1 Representation strategies

Dancers used a range of approaches when designing movement sequences (Table 1). Six of the sequences primarily used body-based or dance terminology—"bend your knees" or "relevé"—to convey movement. Two of the sequences primarily used imagery that was thematically connected, but lacked an overarching narrative, for example describing a variety of interactions with water in different forms. The remaining sequence used a narrative approach that had a clear storyline across all movements in the sequence; for example, imagining a morning routine. Figures 3, 4, and 5 illustrate one movement sequence created for each activity.

Dancers noted data encodings using height in seven movement sequences, effort in six movement sequences, weight in four movement sequences, and direction in three movement sequences. Two dancers described encodings that used emotion as a movement parameter, though dancers had been encouraged to focus on motion-centered encodings. Shape was also observed in two movement sequences. Scale, time, and space were each observed in one movement sequence. In a majority of the movement sequences (7/9) designers described using multiple movement parameters to represent information and in the remaining two sequences movers perceived multiple parameters though designers intended to use only one. This is detailed further in the following section.

4.2 Layering movements and embodied experiences

The intertwined nature of the movement parameters was evident in the design of the movement sequences. In multiple movement sequences (M02, M03, M05, M08) movers commented that they felt effort and weight² varying together, or that the designer could have been using either—or both—of the movement parameters to represent the information. Specifically, dancers felt that a heavier feeling of weight in the body or a feeling of holding the weight lower in the body corresponded to a more effortful movement. While weight and effort may be varied independently in the body, treating them separately may require instructions that specifically

²Our use of the word "weight" throughout this paper aligns with its use in dance and somatics. That is, it refers to the movement quality (e.g. light or strong/heavy), the dancer's relationship to the floor (e.g. level of groundedness), or the way the body or parts of the body are carried and relate to one another. It does not refer to any measurement of kilograms or pounds.

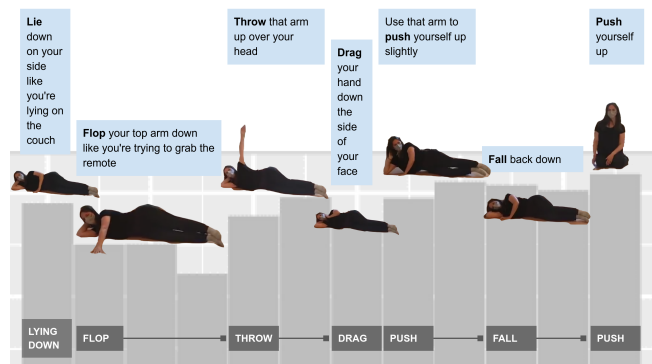


Figure 3: Movement sequence M02 designed by participant P02 for instructional activity #1 takes an imagery-based approach to conveying movement. The designer described two layers of movement parameters representing the data: the use of the body's weight communicates the overall high levels of the data (interpreted as "heaviness") while the changing height and effort in the arm conveys the small variations in the data. The groupings of the data points representing each movement are inconsistent in size and often represent relationships between the points, for example regions of increase or decrease in the data.

indicate that they should vary separately—and may be accessible only to those with considerable somatic training. That is, these movement parameters may be less orthogonal for some people than visual parameters like symbol and color.

Some designers intentionally layered two or more movement parameters to reinforce the data representation. Instructions for M01 incorporated both effort and height throughout the movement sequence; increasing data was represented by movements of increasing effort, and the height of the most active body part of the movement also increased as the data increased (e.g. sliding involves the feet while a plié involves the knees). In visualization language, this might correspond to using both color and symbol to represent one piece of information.

In other cases, different aspects of the data were expressed through different elements of the movement. In one movement sequence (M02) representing data that had consistently high values with small variations (Fig 3), the designer used weight and the whole of the body to express the "heaviness" of the data overall and changes in the height of the hand to express the small variations. In M08, designers represented the changes between points—the slope of the graph—through changes to the direction of movement, and used time (via repetition) to encode the number of points in the segment. In both cases, the overlaid movements gave additional information about the data.

4.3 Movement, feeling, and emotion

These activities and the workshops they were part of centered formal ways of representing data with and through the body in movement. The data, the structure of the workshop, and the instructions for the activities discouraged approaches that aimed to tell a story about the people behind the data or elicit an emotional

response to the data; instead, the activities focused on varying embodied experience through simple movements.

Despite this—and perhaps not surprisingly—dancers remarked on the emotional experience created by the movement sequences. One designer (P01) noted that the movement they selected for the maximum value both followed the representation strategies for the rest of the graph and was selected to create a feeling of excitement (M01). This was also conveyed through the designer's verbal delivery of the instruction ("jump!") which had particular energy around this point. Movers also perceived an emotional layer in one movement sequence that took a narrative approach and described a rushed morning commute (M04).

Another dancer noted that they enjoyed the activities because "it was cool to assign emotion to an inanimate data graph" (P02). They described how in their experience "emotion arose with a concept" using the example of more weight giving the experience of hard emotional situations (P02). They noted that emotion could have been discussed explicitly during the activities, possibly indicating that they found it to be an important part of their experience.

4.4 Surfacing relationships within the data

Dancers' movements and their descriptions of the data, their design processes, and their experiences as movers indicated that they relied on and centered relationships in the data. In particular, data points were often grouped together in the design process and understood primarily relative to each other.

When asked to describe what the data looked like, movers narrated how they perceived the data to be "going up" or "going down" over the arc of the movement sequence and relative to adjacent points; the amounts were less salient than the changes from one part of the data to the next. In instructional activity #1, most of the descriptions of the graphs put forward by the movers broadly matched the shape and direction of the data, though sometimes movers were unsure about the relative amplitudes of different parts of the graph. Perceiving these relationships became more difficult for the movers in instructional activities #2 and #3 as multi-line visualizations were introduced (Figures 4 and 5).

Similarly, designers relied on the relationships in the data in the design process. One noted that "after we looked at the relationships between years, it was easier to come up with movement" (P04) and another discussed using "comparative movements" (P01) in their design process. Designers never referenced the numeric values on the axis when describing their data, though they frequently mentioned considerations around the shape of the graph and the changes between sets of bars ("bigger", "lower"). Designers sometimes made note of extremes in the data that they tried to communicate to the movers and again used relative terms ("biggest", "big spike").

Designers also identified relations between data points when putting together their movement sequences. Eleven of the 12 movement sequences grouped subsets of data points together and represented the subsets through movements, instead of using a movement for each individual data point. In six of the movement sequences (M01-M04, M06, M07), the data points were grouped in non-uniform ways that prioritized relationships between points over standardization; one movement might represent a set of four data points while another movement represented a set of two data

points. For the most part, the differences in set size were not otherwise accounted for—movements representing more data points did not appear to last longer, though one piece of one of the movement sequences indicated that a movement should be done with two hands to represent the two data points encoded in the movement (M01). One of the remaining movement sequences (M05) divided the graph into three movements each conveying four data points and executed over four beats. The other movement sequence (M08) used irregular sized groupings of data points and represented the size of the group through counts used for a particular movement or repetition of that movement.

The sets of data points seemed to be grouped to represent particular kinds of relationships—an area of increase, an area of decrease, an area of largely static data values. In some cases, the boundaries of the sets were fuzzy or overlapping. For example, two "going up" movements might be used for an increasing set of seven data points, without directly assigning three data points to one movement and four data points to the other (M04). In instructional activity #1, the maximum data point was often assigned to its own motion and was not grouped with others, perhaps as a way to emphasize it (M01, M02, M04). Interestingly, in one of these cases (M02), the maximum was not a particularly extreme point of the data (Figure 3).

Designs for multi-line graphs (instructional activities #2 and #3) sometimes used movement to bring out changes in relationships between the lines. Designers of M06 used shape to draw attention to the changing relationships between the two lines (beer sold as kegs and beer sold as cans), specifically how the can line on the graph begins below ("under" movement) but then greatly exceeds ("over" movement) the keg line (Figure 4). This is the only time shape is used in this movement sequence. Similarly, the changing relationships between these lines were central in M07 and expressed through the relative height changes between different parts of the body (the head and chest) throughout the movement sequence.

4.5 Designing through relations in the body

Movers experienced the movement sequences relative to their own bodies. While movers varied in their interpretation of the instructions, there was some consistency in their individual choices. This allowed designers to create movement sequences that surfaced the salient differences even though there was considerable variation in the implementation.

Variation in the interpretations of the movement instructions was particularly noticeable in movement sequences conveyed through narrative or imagery (M02, M03, M04). For example, some movers interpreting the instruction "imagine you're swimming through water on a lake" sat on the floor during the movement, others stood in place, and others walked around the room (M03). Some movers also changed their implementation of the instructions each time they did the movement sequence. Even movement sequences that used dance terminology often left space for movers to make additional decisions; for example, movers interpreting the "plié" instruction in M01 varied in the orientation of their feet, the depth of their knee bend, the speed of their descent, and other movement qualities.

At the same time, movers brought some consistency to choices within any given movement sequence. The choices a mover made about their "plié" in a movement sequence generally held for the

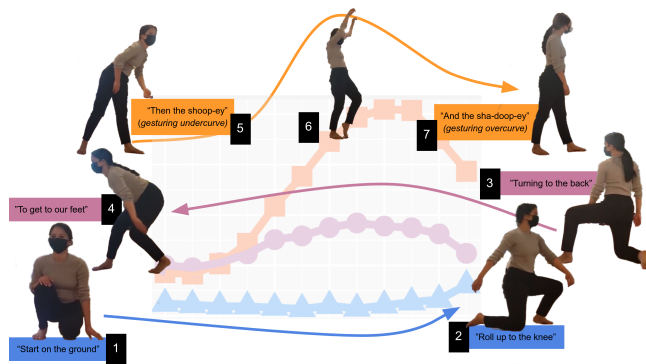


Figure 4: Movement sequence M06 designed by participants P01, P04, and P05 for instructional activity #2 used changes in height to show differences between each line of the graph. Changes in direction indicated the start of a new line, which sometimes moved from right to left and other times from left to right depending on the direction the dancers were facing. Relationships between the orange (beer sold as cans) and purple lines (beer sold as kegs)—particularly the change in relationship when the orange line passes above the purple line—are encoded in the shape of the last three movements.

"relevé" that followed it in the same sequence (M01). When a movement sequence started with the instruction to "lie down on your side like you're lying on the couch," (M02) the original interpretation of that instruction varied between people, but remained consistent during the rest of that sequence. That is, each mover seemed to establish a baseline of movement choices for a given sequence that they maintained unless the instructions told them to alter their movement. This allowed them to feel the relative differences in the movements in their bodies, though each person interpreted the instructions differently.

4.6 Enjoyable challenges with the potential for learning

Overall, dancers had a positive experience with the instructional activities and described a number of potential favorable characteristics or outcomes from the approach. Our evaluation did not prioritize the assessment of these claims, yet these comments from the dancers serve as a preliminary indication that somatically-oriented data representations merit further exploration and likely are of value in some situations.

All dancers commented that they enjoyed the activities. One dancer "really liked this way of thinking/choreographing" (P04) and others described the activities as "fun" or "super fun" (P01, P02, P05). Some dancers noted that they felt the activities provided benefits. One dancer wrote that "I felt like I understood the graphs in a different way" and that "we overall are having more expansive but specific/intentional ways of interpreting the graphs" (P01). Another noted that "figuring out others' graphs based on movement" was challenging, "(but fun challenging)" (P02).

Dancers found the process of designing the movement sequences to be difficult at times, sometimes in a way that required different

kinds of thinking. One dancer described how the instructional approach led them to think more deeply about the representation, noting that in previous workshop activities "it was a linear thought process, but now it felt [like] more effort in thought" partially because they had to "think how [the movements and feelings] applied to my body and also applied to the graph and predict what the graph would look like" (P04). Dancers found the design process for graphs with multiple lines (instructional activities #2 and #3) to be particularly challenging, citing difficulties in negotiating between movement and data (P04), clearly representing changes in the data through movement (P01, P02), and writing instructions that provided sufficient guidance to the movers (P01, P05).

5 DISCUSSION

Many novel representations of data within HCI encode information in objects. In this paper, we explore the possibility of body-first approaches to data representation that prioritize the felt experience through moving and doing. By working with dancers—a group of somatic connoisseurs—we explore the possibilities of embodied data representations through instructional movement activities that take inspiration from the arts. These efforts build from areas within HCI, such as data physicalization, data feminism, and soma design, as a first step to identifying potential directions for somatic embodied data representation.

Overall, we found that dancers were enthusiastic about these approaches to data representation and the potential for future work in this area. Movement sequence designers used narrative, imagery-based, and technical strategies to communicate the data that drew on a variety of movement parameters including height, effort, weight, and direction, among others. The use of emotion in two movement sequences was particularly interesting as the workshop intentionally excluded these considerations and redirected conversations about feeling to "bodily" sensations. Often, parameters were layered in a movement sequence. Layering either reinforced the data—such as when increasing effort and weight were both used to show increasing data—or encoded different aspects of the data, such as when a heavy feeling of weight in the body represented the overall high values in the data and the changing height of the hand showed the small variations in the data. Layered approaches also revealed how our everyday experience of movement parameters may be more intertwined than visual parameters.

Designers also considered data relationally. Most of the movement sequences encoded changes—that is, relationships between sets of points—instead of representing the amounts of individual points. Designers' relational understanding of the body also facilitated the somatic legibility of the movement sequences. Bodies and movements vary considerably from one person to another—some people are taller, others shorter, some people default to moving quickly, others more slowly, each person begins from their own unique posture and somatic awareness. Despite these differences, many of the movement sequences remained interpretable as any individual had some internal consistency from one moment to the next and acted, in a way, as their own relational space—their own framework of axes, scales, and tick marks.

With these findings in mind, we reflect on our work in the context of the broader aims of novel data representations suggested by areas

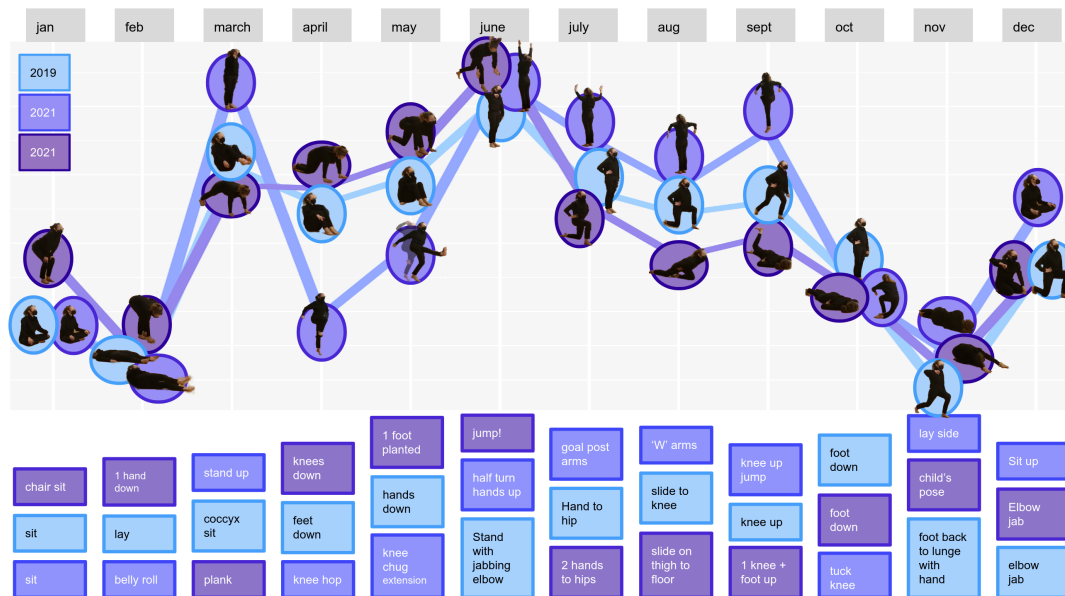


Figure 5: Movement sequence M09 designed by participants P01 and P04 for instructional activity #3. Designers varied the height and scale of the movements to represent the data. This is the only movement sequence that assigns a movement to each data point.

such as data physicalization and data feminism. We also consider the possibility of somatic experiences of data more broadly by returning to soma design and investigating what continued research in this space might include.

5.1 Identifying aims and making meaning

Novel data representations are motivated by a variety of aims, from including more people in conversations about data [47, 48, 51] to supporting learning [9, 25, 36] to making space to acknowledge the emotional aspect of information [48, 62]. Particular data representations may also pursue aesthetic ends [13], justice aims [15], or other outcomes. This exploratory paper does not attempt to fully establish or evaluate how or if somatic data representations might contribute to these or other aims. At the same time, it is worth beginning to reflect on how our work so far might align with or diverge from the broader goals of novel data representations. We reflect in particular on the possibility of expanding the nature of the emotional engagement observed in the movement sequences.

The affective nature of data and the role of emotion in understanding, meaning making, and action provide an area for future development. The topic of the data that we selected for these workshops—beer sales across the United States—was relatively emotionally neutral for the participants in the workshops. When selecting the workshop data, we considered using data on the COVID-19 pandemic and data on environmental challenges but got feedback from some of the dancers that their capacity for engaging with difficult topics was limited in the tumultuous time and that the beer sales data would serve as a relatively lighter option. As researchers, we were also interested in making space for more abstract representations of data—instead of narrative approaches about the larger

issue—which seemed more likely with a topic that was lower stakes for the workshop participants.

At the same time, some dancers brought emotion into their interpretations of the data. These emotional layers focused on the "drama" in the data—the spikes in the graph or the extremeness of changes between points. Emotional sensemaking about the data itself—for example, what it means for people when beer sales double in a year—was not included in the movement sequences and only rarely came up in conversation. For these participants, the emotional meaning of changes to beer sales appeared to be limited relative to other issues on their mind, such as pandemic deaths, environmental disasters, or other emotionally charged topics that we collect data on that informs collective decision making and our day-to-day lives.

Future work could explore what emerges when the stakes around the data are higher for the people designing or participating in the experience. How do designers combine the "drama" within the data and the need for emotional sensemaking around important topics expressed through data? How do motion and emotion get layered into somatic experiences? The Data Sensorium [13] and data theater [7] projects begin to explore these spaces through narrative approaches that use the body to explore data on climate migration and local food security issues. Future work could build on this space by taking a somatics-first approach to data on topics of importance to the participating individuals.

5.2 Somatic experiences of data

This work aimed to facilitate embodied somatic experiences of information. Our early findings indicate that movers often perceived changes in their felt experience and pulled meaning from them.

though this was challenging at times. Dancers also perceived emotion in the movement sequences, though the workshops centered "body" approaches to representing the information. This signals that dancers may have had a holistic experience of embodiment that was integrated across the body and mind—as Hook states, "movement, emotion and thinking are not separable" [31].

Approaches from soma design could support the extension of these efforts to new populations. Our participants were all somatic connoisseurs. Working with less experienced movers and feelers requires further cultivating the soma and—like soma design—specifically identifying this as an aim and outcome of the design process [27]. Our workshops with dancers included warm-up activities that brought people into their bodies and helped them "tune in" to the somatic knowledge they had already nurtured in their movement practices. Workshops with less experienced movers would require more extensive work to build somatic awareness; these practices can take considerable time and effort to develop [44, 60]. This also means that novel data representations that rely on somatic knowledge may not be immediately understood by all people or may be interpreted differently by different groups.

Additionally, work in soma design suggests the possibility of using artefacts to create embodied experiences of information. Our research excluded artefacts from the data representation in order to ensure that movement and the body were the central means of experiencing the data and to diverge from approaches in data physicalization where objects "embody" data. But embodied experiences of information could also be facilitated by objects. Many of Forti's Dance Constructions such as Slate Board [22] and Forsythe's Choreographic Objects [21] suggest particular movement experiences through artefact-centered approaches. Using soma design processes to create data physicalizations provides an opportunity to expand the "material ways" that data physicalizations encode data to include more kinetic, experiential, and holistically embodied interactions with information through objects.

5.3 Exploring the possibilities of data somatics

Based on our work, we propose the area of "data somatics" that centers rich embodied experiences of data that are felt above all else. By using the term somatics, we hope to ensure that body-based approaches will not be overwhelmed by the visually-dominant approaches established in much of the field. We also acknowledge the effort it takes to engage with this type of embodied knowledge—and the potential richness of these data representations.

Our efforts in this paper serve as a first step into this large space of possibility. Based on our initial work, we suggest two areas of exploration that might serve as next steps in this space: (1) further development of a design language for data somatics, (2) re-connecting data somatics to social and community impact.

Through its long history, visualization researchers and practitioners have developed design languages that serve as a common foundation for how we create and interpret data visualizations. Data physicalization³ and data sonification⁴ are each in the process of developing similar shared understandings through collecting examples, finding commonalities, and building a community of practice.

³<http://dataphys.org/list/gallery/>

⁴<https://sonification.design/>

This paper begins to explore a design language of data somatics through collaboration with dancers with modern or contemporary dance backgrounds. Creating more instances of data somatics will expand, refine, and strengthen this starting point. Developing a body of work designed by dancers from various backgrounds, somatic practitioners, designers, and data visualization researchers will shed light on the diverse ways of making sense of movement, bodily experiences, and data. Collaborations across these groups may be particularly fruitful in building toward a design language for data somatics, especially longer term partnerships such as a residency for a Feldenkrais practitioner in an academic visualization lab.

Data somatics draws on ideas from data physicalization and data feminism around increasing access to data, design of data practices that can create positive impact in communities, and the politics of data and the body. An important next step in this work is reconnecting it with those aims through community engaged practices around issues of local importance. This could include workshops that bring together design researchers, somatic practitioners, advocacy organizations, and community members to collaboratively design somatic data experiences. Such a process could create space for collective sensemaking, community building, and advocacy efforts that both speak the language of data and bodily experience—building on models in data murals [9], situated data physicalization design through Participatory Action Research processes [48], and other exploratory data practices. It may also surface and clarify challenges in data somatics such as the variation in somatic awareness among people and the extent to which various communities are enthusiastic about engaging in embodied practices.

6 LIMITATIONS AND FUTURE WORK

This paper is one of the first explorations of somatically-focused ways to create active embodied data representations through movement and begins to help us understand the possibilities in this space. Because of this, the scale is exploratory and small. It intends to sketch out what might be possible and to surface footholds for future work; it does not attempt to create design rules for this space or to systematically measure attributes of the outcomes, such as legibility, learning, or other metrics commonly used to evaluate visualizations. As design knowledge of this space evolves, future work could investigate these areas—or could suggest forms of evaluation specific to somatically-oriented data representations, as it is likely that the aims, values, and design processes in this space will diverge from traditional visualization research.

Our workshop series brought together a small group of dancers, many of whom were part of a dance group led by a member of the study team. This familiarity allowed us to design workshop activities suited to the group's skills, build a comfortable environment for risk taking, and create an open dialogue, though it is also possible that dancers were influenced by the good participant effect. The dancers' shared dance group membership also meant that most of the dancers we worked with were trained primarily in contemporary or modern dance and shared similar language around movement—concepts that we drew on in the workshops. Dancers with different movement backgrounds or somatic practitioners would likely design differently. This provides a rich opportunity for future work.

The dancers in the workshops had a wide variety of experience with data, including some individuals with low comfort with data. As we had limited time, we elected to start the activities from data visualizations instead of data tables and used simple visualizations such as bar graphs or line graphs. This decision took into account the practical realities of our work with this community while, we believe, also creating an opportunity to learn what this emerging space could include. At the same time, it is possible that the form of the visualizations influenced the design outcomes, even though dancers were focused on creating somatic experiences. Future work could begin the design process from raw data or compare movement sequences created from a wider range of visualizations.

7 CONCLUSION

In this paper, we explore somatic movement-based approaches to representing data. By working with dancers, we present an experience of data that is body centered; the data representations exist only in the body—not in artefacts—and are designed to be felt and not viewed. Our findings reveal a number of design strategies including layered and relational approaches. They suggest the potential for future research in data somatics that continues to deepen our ideas about what embodiment includes and expands efforts in HCI to make information more accessible and meaningful.

ACKNOWLEDGMENTS

We thank the dancers who participated in these workshops and helped us explore bodily and movement-based ways of representing data. We also thank Rahul Bhargava, Ilya Vidrin, Victoria Palacin, the Embodiment and Creativity Impact Group, and the broader community of researchers exploring embodied practices in design, data visualization, and the arts. Thanks to Tad Hirsch and Dietmar Offenhuber for helpful discussions and the anonymous TEI reviewers for their thoughtful comments and advice. This work was supported by the department of Art + Design, the Center for Design, and the College of Arts, Media and Design at Northeastern University.

REFERENCES

- [1] Miquel Alfaras, Vasiliki Tsaknaki, Pedro Sanches, Charles Windlin, Muhammad Umair, Corina Sas, and Kristina Höök. 2020. From Biodata to Somadata. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM, Honolulu HI USA, 1–14. <https://doi.org/10.1145/3313831.3376684>
- [2] Yakama Native American. 2018. Yakama Time Ball. <http://dataphys.org/list/yakama-time-ball/>.
- [3] Janine Antoni. 1993. Slumber. <http://www.janineantoni.net/slumber>.
- [4] Birgir Rafn Baldursson, David Peterson, and Mafalda Gamboa. 2022. Nebula: Artistic Somaesthetic Appreciation with Biosignals in Virtual Reality. In *Adjunct Proceedings of the 2022 Nordic Human-Computer Interaction Conference (NordCHI '22)*. Association for Computing Machinery, New York, NY, USA, 1–3. <https://doi.org/10.1145/3547522.3547710>
- [5] Irmgard Bartenieff and Dori Lewis. 1980. *Body Movement: Coping with the Environment* (1st edition ed.). Routledge, New York.
- [6] Janne Mascha Beuthel. 2020. Wearing the Invisible: Wearable Manifestations of Embodied Experiences. In *Companion Publication of the 2020 ACM Designing Interactive Systems Conference (DIS '20 Companion)*. Association for Computing Machinery, New York, NY, USA, 485–489. <https://doi.org/10.1145/3393914.3395828>
- [7] Rahul Bhargava, Amanda Brea, Victoria Palacin, Laura Perovich, and Jesse Hinson. 2022. Data Theatre as an Entry Point to Data Literacy. *Educational Technology & Society* 25, 4 (2022), 93–108.
- [8] Rahul Bhargava and Catherine D'Ignazio. 2017. Data Sculptures as a Playful and Low-Tech Introduction to Working with Data. *Presented at Designing Interactive Systems, Edinburgh, Scotland* (2017).
- [9] Rahul Bhargava, Ricardo Kadouaki, Emily Bhargava, Guilherme Castro, and Catherine D'Ignazio. 2016. Data Murals: Using the Arts to Build Data Literacy. *The Journal of Community Informatics* 12, 3 (2016).
- [10] Anne Bogart and Tina Landau. 2004. *The Viewpoints Book: A Practical Guide to Viewpoints and Composition*. Theatre Communications Group.
- [11] Sophia Brueckner and Rachel Freire. 2018. Embodisuit: A Wearable Platform for Embodied Knowledge. In *Proceedings of the Twelfth International Conference on Tangible, Embedded, and Embodied Interaction*. 542–548.
- [12] Nadia Campo Woytuk, Marie Louise Juul Søndergaard, Marianela Ciolfi Felice, and Madeline Balaam. 2020. Touching and Being in Touch with the Menstruating Body. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–14.
- [13] Lins Derry. 2023. Data Embodiment: Approaching the Body as a Choreographic Medium for Performing Abstract Data. *International Journal of Performance Arts and Digital Media* 19, 1 (2023), 60–82.
- [14] Audrey Desjardins, Oscar Tomico, Andrés Lucero, Marta E. Cecchinato, and Carman Neustaedter. 2021. Introduction to the Special Issue on First-Person Methods in HCI. 12 pages.
- [15] Catherine D'Ignazio and Lauren F. Klein. 2020. *Data Feminism*. MIT Press.
- [16] Kelly Dobson. 2016. The Energy Propagated. <https://cargocollective.com/TheEnergyPropagated>.
- [17] Paul Dourish. 2001. *Where the Action Is*. MIT press Cambridge.
- [18] Paul Dourish. 2013. Epilogue: Where the Action Was, Wasn't, Should Have Been, and Might yet Be. *ACM Transactions on Computer-Human Interaction (TOCHI)* 20, 1 (2013), 1–4.
- [19] Martha Eddy. 2016. *Mindful Movement: The Evolution of the Somatic Arts and Conscious Action*. Intellect Books.
- [20] Moshe Feldenkrais. 2009. *Awareness Through Movement: Easy-to-Do Health Exercises to Improve Your Posture, Vision, Imagination, and Personal Awareness* (reprint edition ed.). HarperOne, San Francisco.
- [21] William Forsythe. 2022. Choreographic Objects. www.williamforsythe.com/installations.html.
- [22] Simone Forti. 1961. Simone Forti. Slant Board. 1961 | MoMA. <https://www.moma.org/collection/works/200115>.
- [23] Mafalda Gamboa, Mehmet Aydın Baytaş, Sjoerd Hendriks, and Sara Ljungblad. 2023. Wisp: Drones as Companions for Breathing. In *Proceedings of the Seventeenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '23)*. Association for Computing Machinery, New York, NY, USA, 1–16. <https://doi.org/10.1145/3569009.3572740>
- [24] Peggy Hackney. 2000. *Making Connections: Total Body Integration Through Bartenieff Fundamentals* (1st edition ed.). Routledge, New York (N.Y.).
- [25] Trevor Hogan, Uta Hinrichs, Yvonne Jansen, Samuel Huron, Pauline Gourlet, Eva Hornecker, and Bettina Nissen. 2017. Pedagogy & Physicalization: Designing Learning Activities around Physical Data Representations. In *Proceedings of the 2017 ACM Conference Companion Publication on Designing Interactive Systems*. 345–347.
- [26] Trevor Hogan and Eva Hornecker. 2013. In Touch with Space: Embodying Live Data for Tangible Interaction. In *Proceedings of the 7th International Conference on Tangible, Embedded and Embodied Interaction*. 275–278.
- [27] Kristina Höök. 2018. *Designing with the Body: Somaesthetic Interaction Design*. MIT Press.
- [28] Kristina Höök. 2020. Soma Design-Intertwining Aesthetics, Ethics and Movement. In *Proceedings of the Fourteenth International Conference on Tangible, Embedded, and Embodied Interaction*. 1–1.
- [29] Kristina Höök, Steve Benford, Paul Tennent, Vasiliki Tsaknaki, Miquel Alfaras, Juan Martinez Avila, Christine Li, Joseph Marshall, Claudia Dauden Roquet, and Pedro Sanches. 2021. Unpacking Non-Dualistic Design: The Soma Design Case. *ACM Transactions on Computer-Human Interaction (TOCHI)* 28, 6 (2021), 1–36.
- [30] Kristina Höök, Baptiste Caramiaux, Cumhur Erkut, Jodi Forlizzi, Nassrin Hajinejad, Michael Haller, Caroline CM Hummels, Katherine Isbister, Martin Jonsson, and George Khut. 2018. Embracing First-Person Perspectives in Soma-Based Design. In *Informatics*, Vol. 5. MDPI, 8.
- [31] Kristina Höök, Sara Eriksson, Marie Louise Juul Søndergaard, Marianela Ciolfi Felice, Nadia Campo Woytuk, Ozgun Kilic Afsar, Vasiliki Tsaknaki, and Anna Ståhl. 2019. Soma Design and Politics of the Body. In *Proceedings of the Halfway to the Future Symposium 2019*. 1–8.
- [32] Kristina Höök, Martin P. Jonsson, Anna Ståhl, and Johanna Mercurio. 2016. Somaesthetic Appreciation Design. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. 3131–3142.
- [33] Noura Howell, John Chuang, Abigail De Kosnik, Greg Niemeyer, and Kimiko Ryokai. 2018. Emotional Biosensing: Exploring Critical Alternatives. *Proceedings of the ACM on Human-Computer Interaction* 2, CSCW (Nov. 2018), 1–25. <https://doi.org/10.1145/3274338>
- [34] Dianya Mia Hua, Jeffrey Bardzell, and Shaowen Bardzell. 2023. Embodied Embroidery: Somaesthetic Interaction Design for Women's Masturbation. In *Proceedings of the Seventeenth International Conference on Tangible, Embedded, and Embodied Interaction*. 1–15.

- [35] Sarah Hughes, John Collins, Mark Hansen, Ben Rubin, and Jer Thorp. 2015. A Sort of Joy (Thousands of Exhausted Things). <https://www.sarahcameronhughes.com/a-sort-of-joy>.
- [36] Samuel Huron, Benjamin Bach, Uta Hinrichs, Mandy Keck, and Jonathan Roberts. 2020. IEEE VIS Workshop on Data Vis Activities to Facilitate Learning, Reflecting, Discussing, and Designing. In *IEEE VIS 2020*.
- [37] Jörn Hurtienne. 2018. Possibilities of Human Data Embodiment: 100% City. In *Position Paper for the Workshop: Towards a Design Language for Data Physicalization at IEEE VIS*, Vol. 6.
- [38] Yvonne Jansen and Pierre Dragicevic. 2018. List of Physical Visualizations.
- [39] Yvonne Jansen, Pierre Dragicevic, and Jean-Daniel Fekete. 2013. Evaluating the Efficiency of Physical Visualizations. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 2593–2602.
- [40] Yvonne Jansen, Pierre Dragicevic, Petra Isenberg, Jason Alexander, Abhijit Karnik, Johan Kildal, Sriram Subramanian, and Kasper Hornbæk. 2015. Opportunities and Challenges for Data Physicalization. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. 3227.
- [41] Lee Jones, Greta Grip, and Sara Nabil. 2023. Wear Your Heart on Your Sleeve: Using Digital Knitting Machines to Craft Wearable Biodata Portraits. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference (DIS '23)*. Association for Computing Machinery, New York, NY, USA, 547–563. <https://doi.org/10.1145/3563657.3596007>
- [42] Maria Karyda, Danielle Wilde, and Mette Gislev Kjærsgaard. 2020. Narrative Physicalization: Supporting Interactive Engagement with Personal Data. *IEEE Computer Graphics and Applications* 41, 1 (2020), 74–86.
- [43] Bongshin Lee, Petra Isenberg, Nathalie Henry Riche, and Sheelagh Carpendale. 2012. Beyond Mouse and Keyboard: Expanding Design Considerations for Information Visualization Interactions. *IEEE Transactions on Visualization and Computer Graphics* 18, 12 (2012), 2689–2698.
- [44] Wonjun Lee, Youn-kyung Lim, and Richard Shusterman. 2014. Practicing Somaesthetics: Exploring Its Impact on Interactive Product Design Ideation. In *Proceedings of the 2014 Conference on Designing Interactive Systems*. 1055–1064.
- [45] Lian Loke and Thecla Schiphorst. 2018. The Somatic Turn in Human-Computer Interaction. *Interactions* 25, 5 (Aug. 2018), 54–5863. <https://doi.org/10.1145/3236675>
- [46] Claire O'Malley and Danae Stanton Fraser. 2004. Literature Review in Learning with Tangible Technologies. (2004).
- [47] Laura J. Perovich, Phoebe Cai, Amber Guo, Kristin Zimmerman, Katherine Pashman, Dayanna Espinoza Silva, and Julia G. Brody. 2020. Data Clothing and BigBarChart: Designing Physical Data Reports on Indoor Pollutants for Individuals and Communities. *IEEE Computer Graphics & Applications* 41, 1 (2020), 87.
- [48] Laura J. Perovich, Sara Ann Wylie, and Roseann Bongiovanni. 2020. Chemicals in the Creek: Designing a Situated Data Physicalization of Open Government Data with the Community. *IEEE Transactions on Visualization and Computer Graphics* (2020).
- [49] Aura Pon, Eric Pattison, Lawrence Fyfe, Laurie Radford, and Sheelagh Carpendale. 2017. Torrent: Integrating Embodiment, Physicalization and Musification in Music-Making. In *Proceedings of the Eleventh International Conference on Tangible, Embedded, and Embodied Interaction*. 209–216.
- [50] Annina Rüst. 2017. A Piece of the Pie Chart.
- [51] Kim Sauvé, Pierre Dragicevic, and Yvonne Jansen. 2023. Edo: A Participatory Data Physicalization on the Climate Impact of Dietary Choices. In *Proceedings of the Seventeenth International Conference on Tangible, Embedded, and Embodied Interaction*. 1–13.
- [52] Thecla Schiphorst. 2011. Self-Evidence: Applying Somatic Connoisseurship to Experience Design. In *CHI'11 Extended Abstracts on Human Factors in Computing Systems*. 145–160.
- [53] Orit Shaer and Eva Hornecker. 2010. *Tangible User Interfaces: Past, Present, and Future Directions*. Now Publishers Inc.
- [54] Lawrence Shapiro. 2019. *Embodied Cognition* (2nd edition ed.). Routledge, London ; New York.
- [55] Anna Ståhl, Martin Jonsson, Johanna Mercurio, Anna Karlsson, Kristina Höök, and Eva-Carin Banka Johnson. 2016. The Soma Mat and Breathing Light. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems*. 305–308.
- [56] Anna Ståhl, Vasiliki Tsaknaki, and Madeline Balaam. 2021. Validity and Rigour in Soma Design-Sketching with the Soma. *ACM Transactions on Computer-Human Interaction (TOCHI)* 28, 6 (2021), 1–36.
- [57] Simon Stusak, Moritz Hobe, and Andreas Butz. 2016. If Your Mind Can Grasp It, Your Hands Will Help. In *Proceedings of the TEI'16: Tenth International Conference on Tangible, Embedded, and Embodied Interaction*. 92–99.
- [58] Dag Svanaes and Martin Solheim. 2016. Wag Your Tail and Flap Your Ears: The Kinesthetic User Experience of Extending Your Body. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems*. 3778–3779.
- [59] Paul Tennent, Joe Marshall, Vasiliki Tsaknaki, Charles Windlin, Kristina Höök, and Miquel Alfaras. 2020. Soma Design and Sensory Misalignment. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–12.
- [60] Vasiliki Tsaknaki, Madeline Balaam, Anna Ståhl, Pedro Sanches, Charles Windlin, Pavel Karpashevich, and Kristina Höök. 2019. Teaching Soma Design. In *Proceedings of the 2019 on Designing Interactive Systems Conference*. 1237.
- [61] Vasiliki Tsaknaki, Kelsey Cotton, Pavel Karpashevich, and Pedro Sanches. 2021. “Feeling the Sensor Feeling You”: A Soma Design Exploration on Sensing Non-habitual Breathing. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA, 1–16. <https://doi.org/10.1145/3411764.3445628>
- [62] Yun Wang, Adrien Segal, Roberta Klatzky, Daniel F. Keefe, Petra Isenberg, Jörn Hurtienne, Eva Hornecker, Tim Dwyer, and Stephen Barrass. 2019. An Emotional Response to the Value of Visualization. *IEEE computer graphics and applications* 39, 5 (2019), 8–17.
- [63] Jack Zhao and Andrew Vande Moere. 2008. Embodiment in Data Sculpture: A Model of the Physical Visualization of Information. In *Proceedings of the 3rd International Conference on Digital Interactive Media in Entertainment and Arts*. ACM, 343–350.